

Fundamental Physics of Semiconductors

First, we need to analyze the difference in the structure of different materials.

- **Conductors:** Materials that have a loosely bound valence electrons (No Band Gap).
- **Semiconductors:** Materials that have a smaller Band Gap compared to insulators.
- **Insulators:** Materials that have a tightly bound valence electrons (Huge Band Gap).

Characteristics	Conductor	Semiconductor	Insulator
Conductivity	High	Moderate	Low
Resistivity	Low	Moderate	Very high
Forbidden gap	No forbidden gap	Small forbidden gap	Large forbidden gap
Temperature coefficient	Positive	Negative	Negative
Conduction	Large number of electrons for conduction	Very small number of electrons for conduction	Moderate number of electrons for conduction
Conductivity value	Very high 10^{-7} mho/m	Between those of conductors and insulators, i.e., 10^{-7} mho/m to 10^{-13} mho/m	Negligible like 10^{-13} mho/m
Resistivity value	Negligible; less than 10^{-5} Ω -m	Between those of conductors and insulators, i.e., 10^5 Ω -m to 10^8 Ω -m	Very high; more than 10^5 Ω -m
Current flow	Due to free electrons	Due to holes and free electrons	Due to negligible free electrons
Number of carriers at normal temperature	Very high	Low	Negligible
Band overlap	Both conduction and valence bands are overlapped	Both bands are separated by an energy gap of 1.1 eV	Both bands are separated by an energy gap of 6–10 eV
Zero Kelvin behavior	Acts like a superconductor	Acts like an insulator	Acts like an insulator
Formation	Formed by metallic bonding	Formed by covalent bonding	Formed by ionic bonding
Valence electrons	One valence electron in outermost shell	Four valence electrons in outermost shell	Eight valence electrons in outermost shell
Examples	Copper, mercury, aluminum, silver	Germanium, silicon	Wood, rubber, mica, paper

A comparison between different materials

What are Energy Bands?

There are two types of energy bands, valence and conduction bands. The difference between them is the energy each one contains as the conduction bands have higher energy than valence bands.

What effect do **band gaps** have on **conductivity**?



1. Conductors 2. Semiconductors 3. Insulators

From the figure above, we can see that for conduction to happen, electrons need to move from valence bands to conduction bands, therefore, band gaps determine material conduction properties.

We can find that:

For **Conductors**: There is no band gap, allowing the electrons to flow from valence to conduction band, without needing **energy**.

For **insulators**: The band gap is huge, making the flow of electrons harder than conductors, and therefore requiring more **energy**.

For **semiconductors**: The band gap isn't as big as the insulators', making the flow of electrons less hard than insulators, yet harder than conductors, requiring less **energy** for the electrons to flow than in insulators.

From this analysis, we can conclude that **the atomic structure** determines the level of **conductivity** of the material.

How?

Band Gaps $\longrightarrow$ **Energy** $\longrightarrow$ **Less Conductivity**

(The more energy is required, the less conductivity the material is)

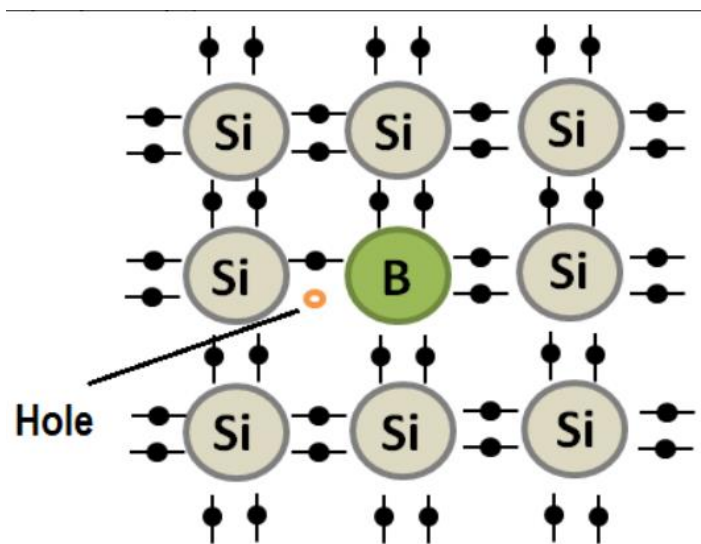
What are **Charge Carriers**?

Charge Carriers are the particles that **carry electric charge** and move freely within the semiconductor material.

There are two types of charge carriers:

Electrons: **Negatively** charged particles that orbit around the nucleus.

Holes: Represent the **absence** of electrons in an atom, has a charge that is equal in magnitude but opposite in polarity to the charge of electrons (**positively** charged).

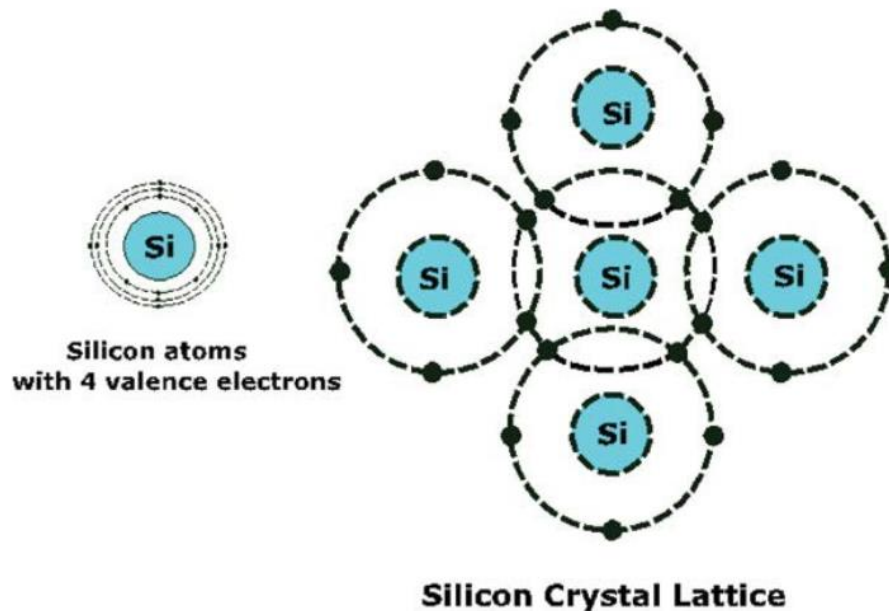


The figure shows that a hole is the absence of an electron

What are **intrinsic and extrinsic semiconductors**?

intrinsic semiconductors are the **pure** materials, in other words, free from impurities, while extrinsic semiconductors are semiconductors that are **doped** by an impurity, that can change the electrical properties of the pure semiconductor.

Intrinsic Semiconductor:



(The silicon used as an example of a pure semiconductor)

Extrinsic Semiconductor:

The behavior of extrinsic semiconductors is determined by impurities, there are two types of extrinsic semiconductors:

- **N-Type semiconductors (Has more Negative charge carriers)**
- **P-Type semiconductors (Has more Positive charge carriers)**

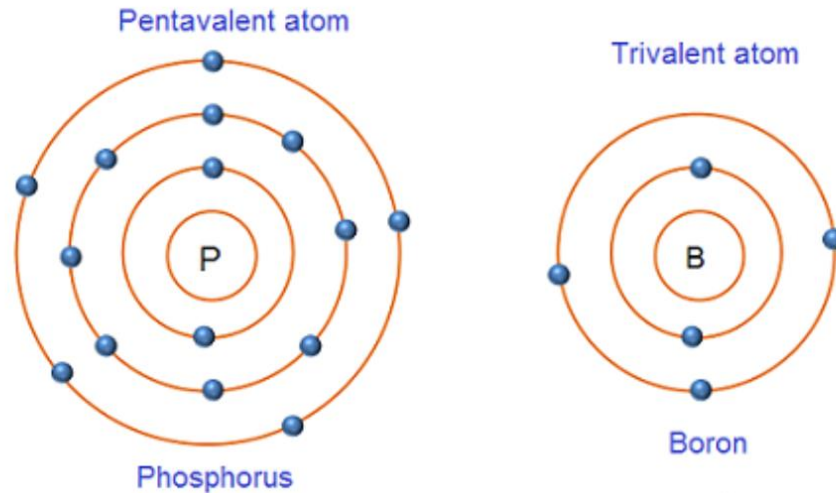
To understand each type, we need to first understand what doping is.

What is **doping?**

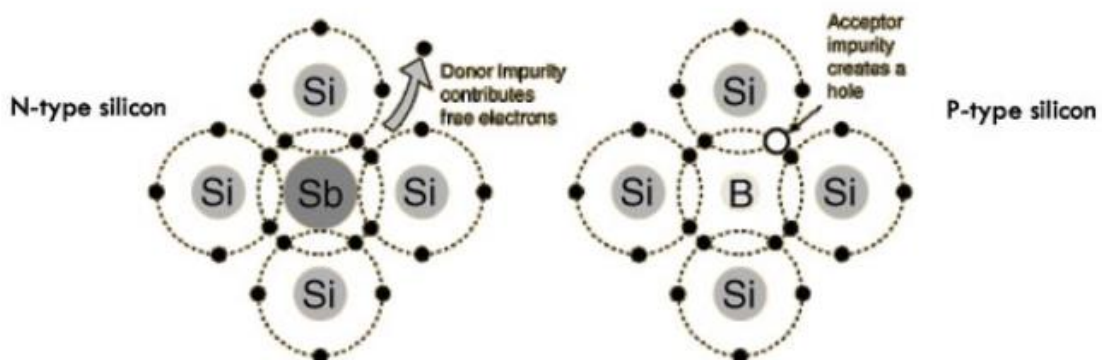
Doping is the process of adding an impurity (or more) to a semiconductor to change its electrical properties by increasing its ability to conduct electricity.

There are two types of impurities:

- Trivalent (**Ex. Boron**)
- Pentavalent (**Ex. Phosphorus**)



The semiconductor type depends on the impurity added to it, if it is **trivalent** atoms the resultant semiconductor is **P-Type**, if it is **pentavalent** atoms the resultant semiconductor is **N-Type**.



The figure shows the difference between N-Type and P-Type semiconductors

has **unique requirements**, so engineers must select materials based on **specific properties** such as:

- **Electron mobility**
- **Band gap**
- **Flexibility**
- **Chemical stability**
- **Thermal conductivity**

Examples of Applications and Material Selection::

1-Transistors (used in processors, memory, logic circuits):



Material Choices:

- **Silicon (Si):** Industry standard due to good performance and low cost.
- **MoS₂ (Molybdenum disulfide):** 2D material with a natural band gap and excellent scaling potential.

2-Photodetectors / Solar Cells:

- **Material Choices:**
 - Gallium Arsenide (GaAs): High efficiency for solar conversion.
 - Quantum dots (e.g., CdSe): Adjustable band gap based on particle size.

3-Flexible Electronics (used in wearable tech, foldable screens):

Material Choices:

- **Graphene:** Extremely strong and flexible with excellent conductivity.

- Organic semiconductors: Low-cost and printable on flexible surfaces.

4-Sensors (gas sensors, biosensors):

- **Material Choices:**
 - **Graphene and MoS₂:** Large surface area, sensitive to molecule interactions
 - **Carbon Nanotubes:** Excellent for detecting gases due to high surface-to-volume ratio.



Methane Gas Sensor

nanodevice applications:

? **Miniaturization of Devices:**

- Modern electronics (like laptops, phones, etc.) contain **billions of silicon devices** as small as **5 nanometers** (about 20 atoms wide).
- Device dimensions have shrunk **by an order of magnitude** over the past decade.

? **Advanced Fabrication Techniques:**

- These devices are manufactured using **deep ultraviolet (DUV) lithography** for precise patterning at the nanoscale.

? **Performance Enhancements:**

- New technologies introduced to improve performance:
 - **3D device architectures**
 - **High-k dielectric materials**
 - **High-mobility materials**

- All of these rely on **careful material selection**.

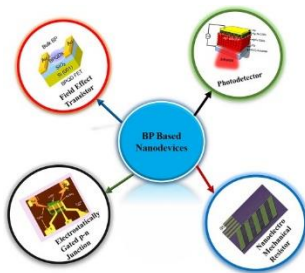
? **Quantum Challenges:**

At very small sizes, **quantum effects** appear:

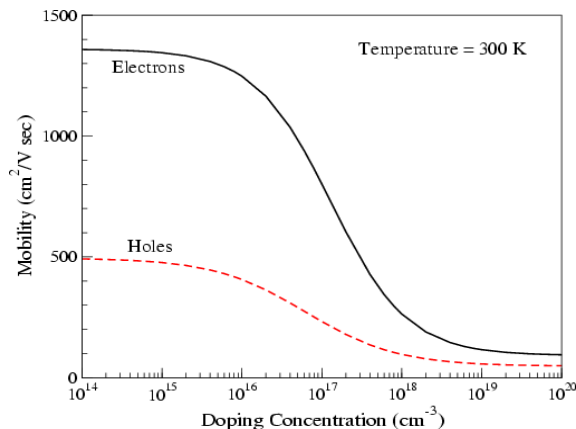
- **Quantum tunneling**
- **Small number of electrons per device**
- Material choice becomes crucial to **control or utilize these effects**.

❓ **Role of Research:**

- Researchers aim to **understand, embrace, and exploit quantum effects** using new **nanomaterials and device designs**.



• **Compare material properties relevant to nanoscale performance:**



Mobility :: is a measure of the ability of an electron or a hole to get around inside a semiconductor. It is a “constant” for a particular situation—a “situation” meaning a particular temperature and doping concentration.

Band gap : The energy required for electrons and holes to transition from the valence band to the conduction band is called a band gap. A semiconductor with a large value is called a wide-band-gap semiconductor. They have a **moderate band gap**, which allows controlled conductivity — making them perfect for electronic devices.

Examples:

- **Silicon:**
Band Gap $\approx 1.1 \text{ eV}$ $\rightarrow$ Ideal for making transistors and circuits.
- **Germanium:**
Band Gap $\approx 0.66 \text{ eV}$ $\rightarrow$ Conducts more easily than silicon.

Flexibility:

- **What is it?**
The ability of a material to bend or deform without breaking.
- **Why is it important?**
It's essential in **flexible electronics**, **wearables**, and **foldable displays**.

At the nanoscale:
Materials such as **carbon nanotubes** and **graphene** combine **flexibility with high strength**, which is a rare and valuable combination.

Property	Nanomaterial Example	Nanoscale Performance
Electron Mobility	Graphene	Extremely high mobility
Band Gap	Quantum Dots	Tunable by changing particle size
Flexibility	Carbon Nanotubes	Highly flexible and mechanically strong

Fabrication Methods for Semiconductor Nanodevices

1. Advanced Lithography Techniques

Lithography is one of the most important industrial methods for nano fabrication in the modern era.

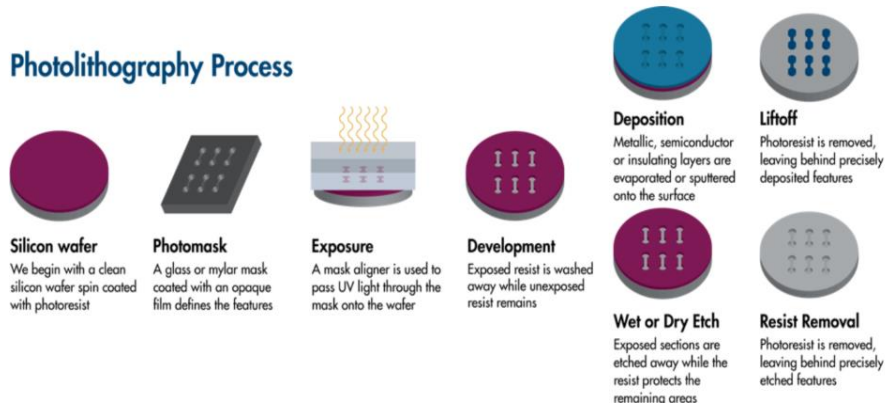
Photolithography

- **Description:** Uses light to transfer patterns from a photomask to a light-sensitive photoresist on the wafer. The photoresist is exposed, developed, and used as a mask for etching or deposition.
- **Process:**
 1. Coat wafer with photoresist.
 2. Expose photoresist to UV light through a mask.
 3. Develop photoresist to reveal the pattern.

4. Etch or deposit material, then remove photoresist.

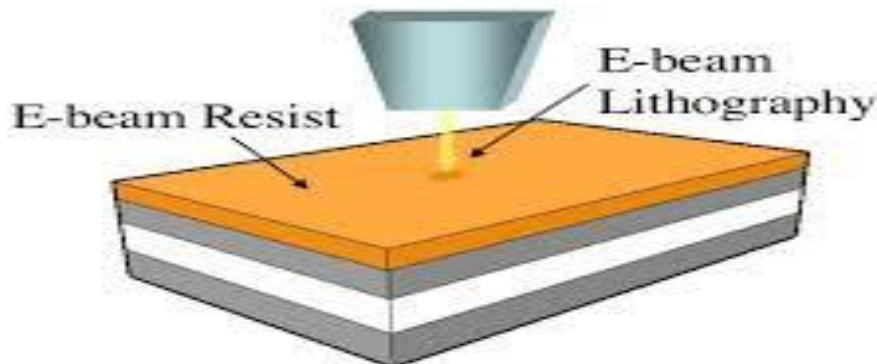
- **Advantages:** High throughput, cost-effective for large-scale production.
- **Limitations:** Limited by diffraction to ~ 193 nm (using deep ultraviolet, DUV). Extreme ultraviolet (EUV) lithography extends resolution to ~ 2 nm but is costly.

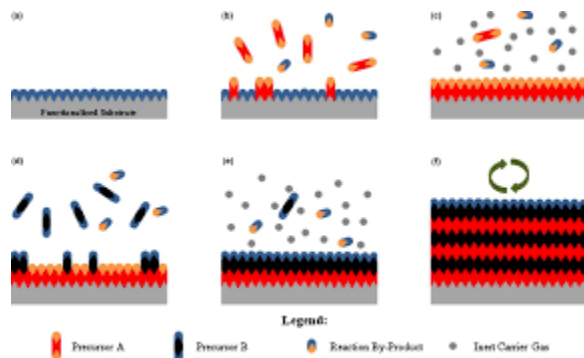
Photolithography Process



E-Beam Lithography

- **Description:** Uses a focused electron beam to directly write patterns onto a resist-coated wafer, eliminating the need for a mask.
- **Process:**
 1. Coat wafer with electron-sensitive resist.
 2. Scan electron beam to expose resist in desired pattern.
 3. Develop resist and proceed with etching or deposition.
- **Advantages:** High resolution (< 10 nm), ideal for prototyping and research.
- **Limitations:** Slow, low throughput, expensive for mass production.





2. Nanoscale Doping and Layering Methods

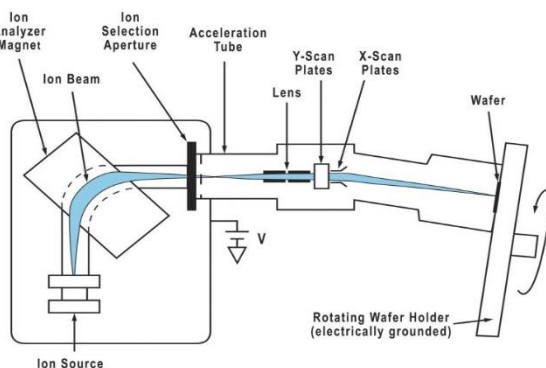
Doping

- **Description:** Doping adds impurities to change conductivity, often with ion implantation for accuracy, and layering builds thin films, with ALD offering atomic-level

control.

- **Ion Implantation:**
 - High-energy ions (e.g., boron, phosphorus) are accelerated and implanted into the substrate.
 - **Advantages:** Precise control of dopant depth and concentration.
 - **Challenges:** Lattice damage requires annealing to repair.
- **Atomic Layer Doping:**
 - Uses atomic layer deposition (ALD) to deposit dopants layer-by-layer with atomic precision.
 - **Advantages:** Uniform doping for ultra-thin layers.
 - **Challenges:** Slow process, limited to specific materials.

Layering



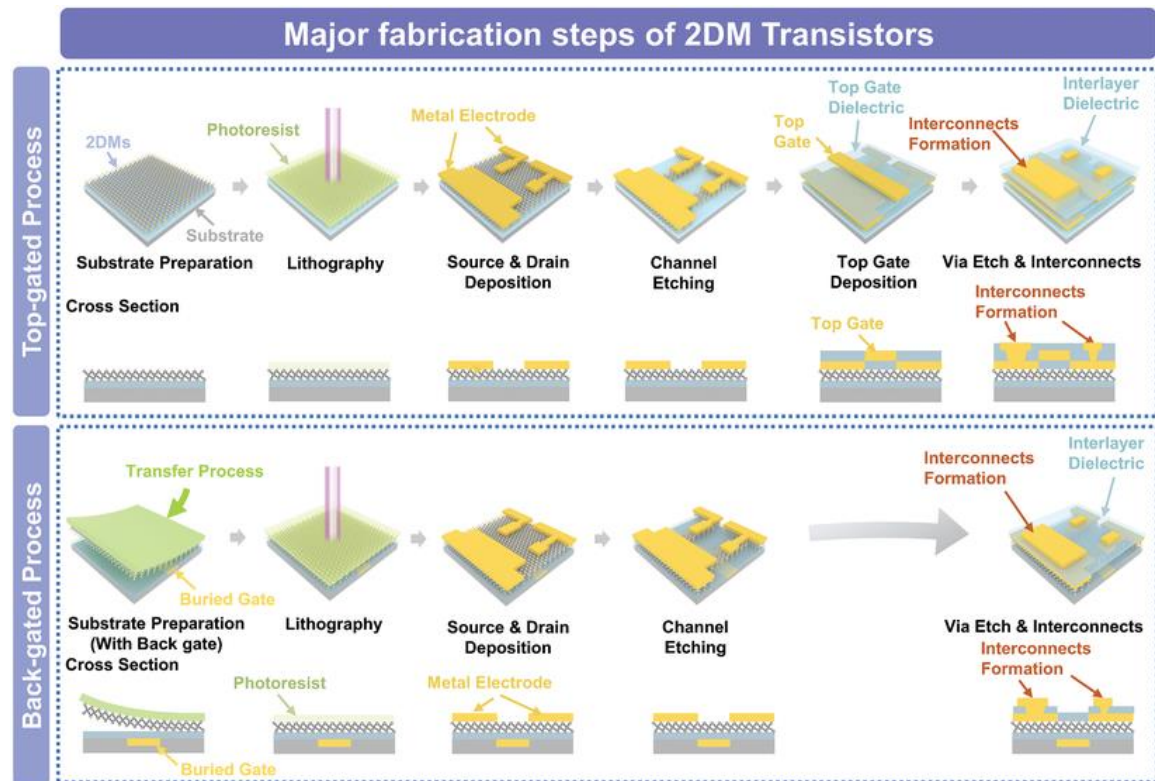
Layering builds the thin films required for nanodevices.

- **Atomic Layer Deposition (ALD):**
 - Deposits materials one atomic layer at a time via sequential chemical reactions.
 - **Advantages:** Excellent uniformity, precise thickness control (~0.1 nm).
- **Chemical Vapor Deposition (CVD):**
 - Gases react on the wafer surface to form thin films.
 - **Advantages:** High deposition rates, suitable for large-scale production.
 - **Challenges:** Less precise than ALD for ultra-thin films.

3. Main Steps in Manufacturing a Nanoscale Transistor

The fabrication of a nanoscale transistor (e.g ; FinFET) involves multiple steps:

1. Substrate Preparation:



- Start with a silicon wafer, cleaned and polished.

2. Fin Formation:

- Use photolithography and etching to create 3D silicon fins.

3. Gate Dielectric Deposition:

- Deposit a high-k dielectric (e.g; HfO_2) using ALD for insulation.

4. Gate Electrode Formation:

- Deposit metal gate (e.g; TiN) using CVD or sputtering.

5. Source/Drain Doping:

- Implant dopants to form source and drain regions.

6. Spacer Formation:

- Deposit insulating spacers (e.g; SiN) to isolate gate from source/drain.

7. Interconnects:

- Deposit metal layers (e.g; copper) for electrical connections.

8. Testing and Packaging:

- Test functionality and package the chip.

4. Challenges in Nanoscale Fabrication

- **Precision:**
 - Achieving sub-10 nm features requires extreme accuracy. EUV lithography and e-beam lithography are costly and complex.
- **Defects:**
 - Atomic-scale defects (e.g; vacancies, dislocations) can degrade performance. Contamination control is critical.
- **Cost:**
 - Advanced tools (e.g; EUV lithography machines) cost hundreds of millions of dollars. R&D for new processes is expensive.
- **Scalability:**
 - Balancing high precision with high throughput remains a challenge.

References:

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